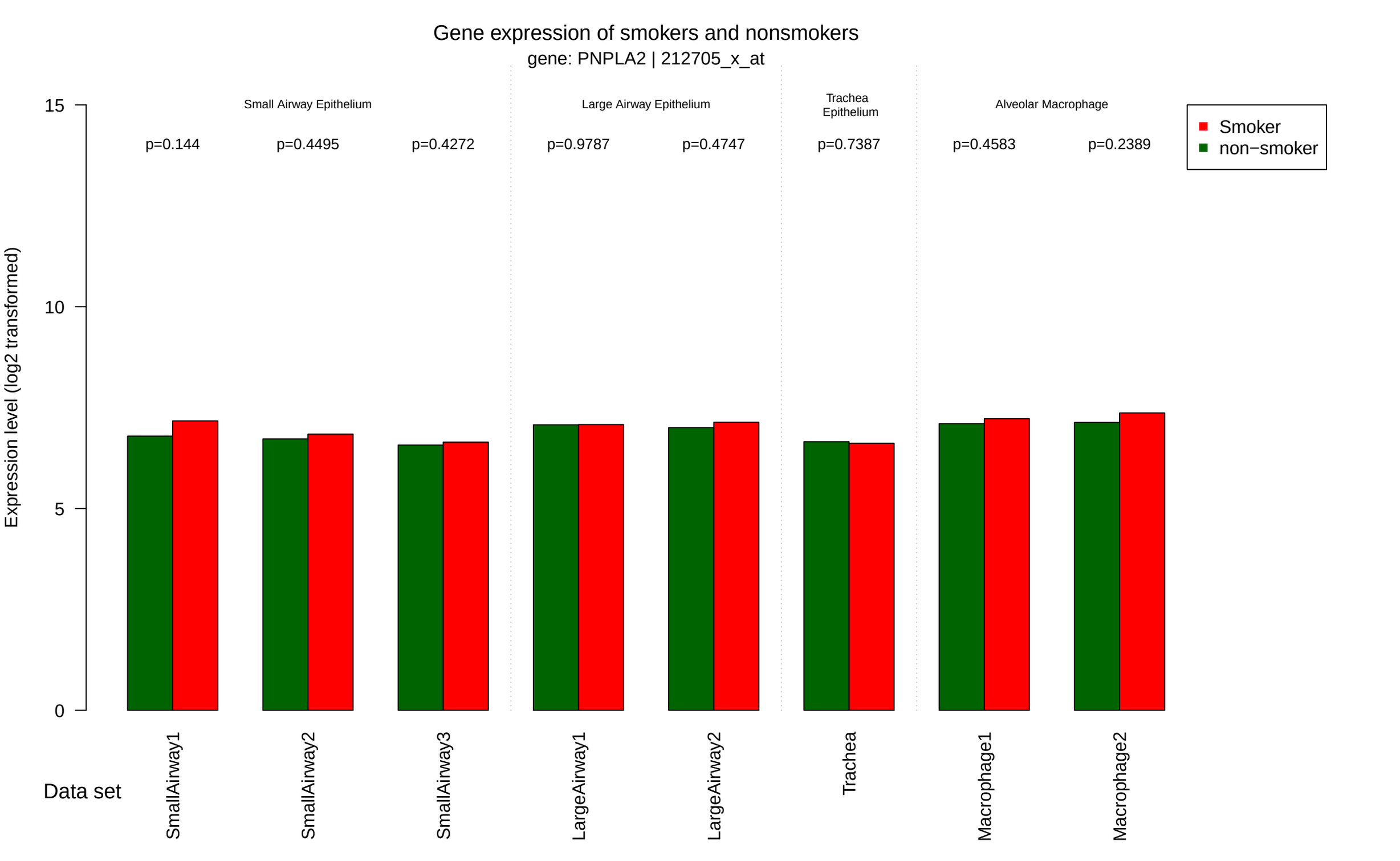
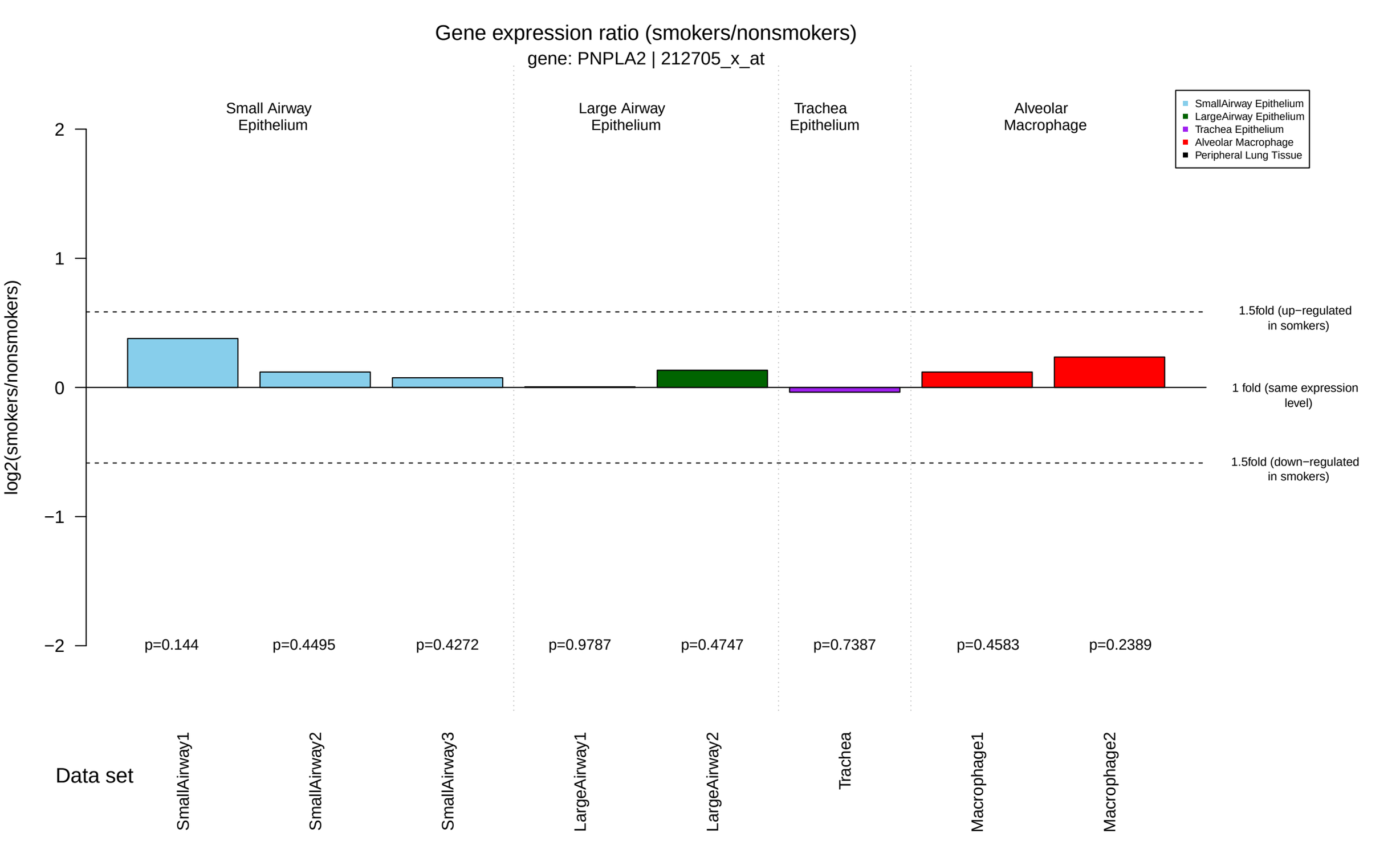
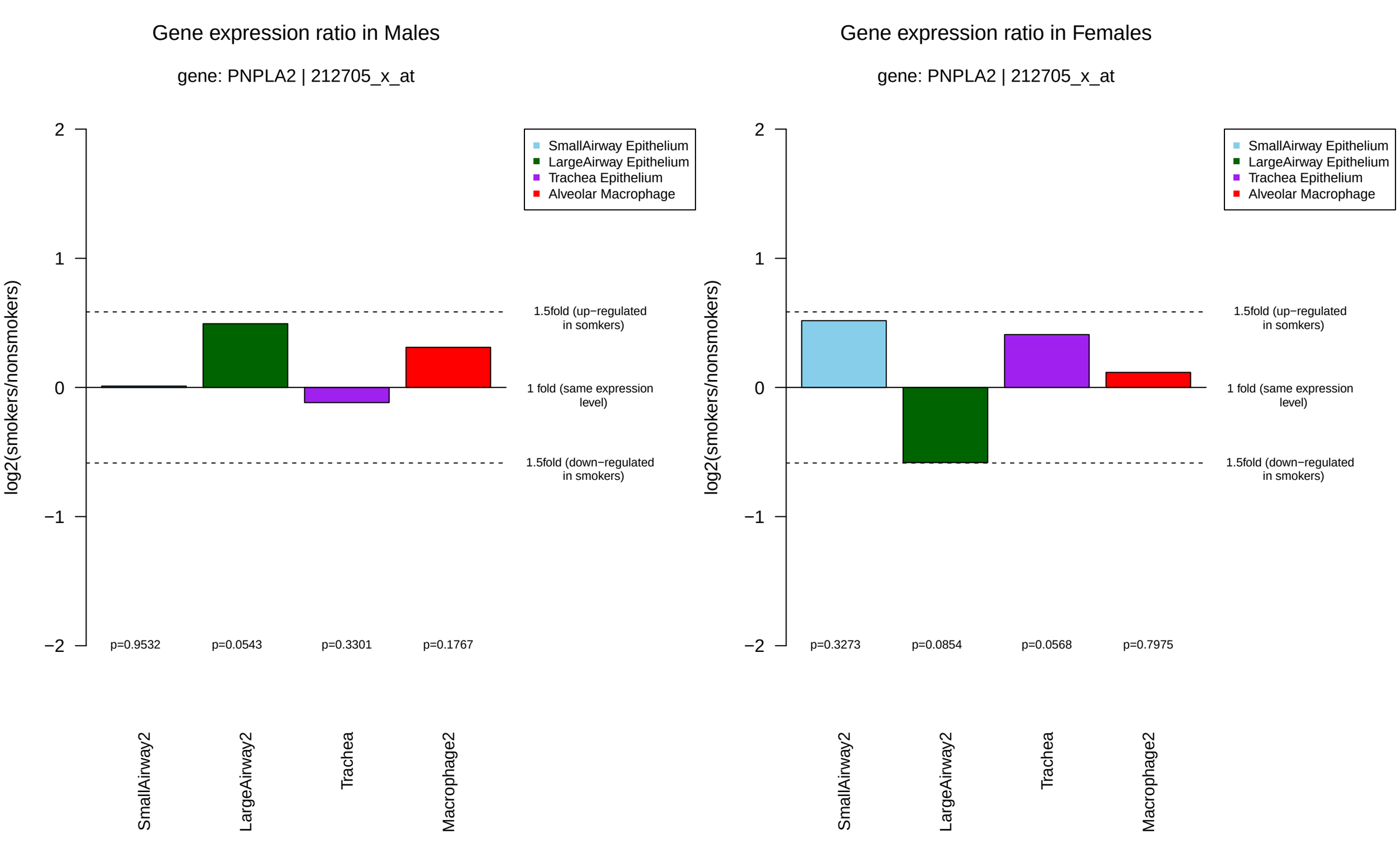
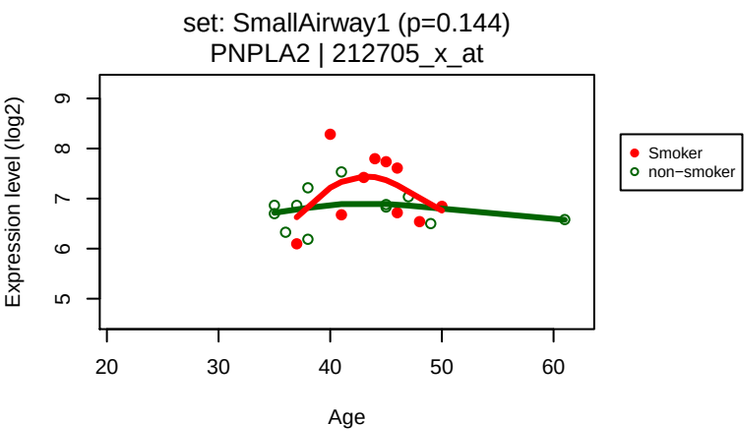
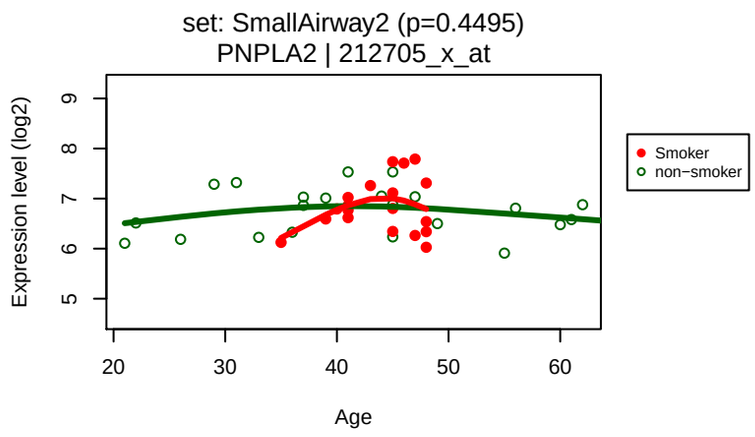
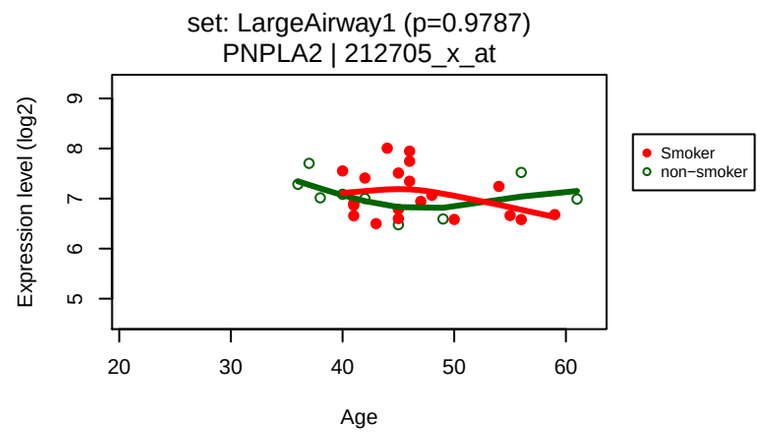

Gene expression of smokers and nonsmokers
gene: PNPLA2 | 212705_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.144
p=0.4495
p=0.4272
p=0.9787
p=0.4747
p=0.7387
p=0.4583
p=0.2389
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

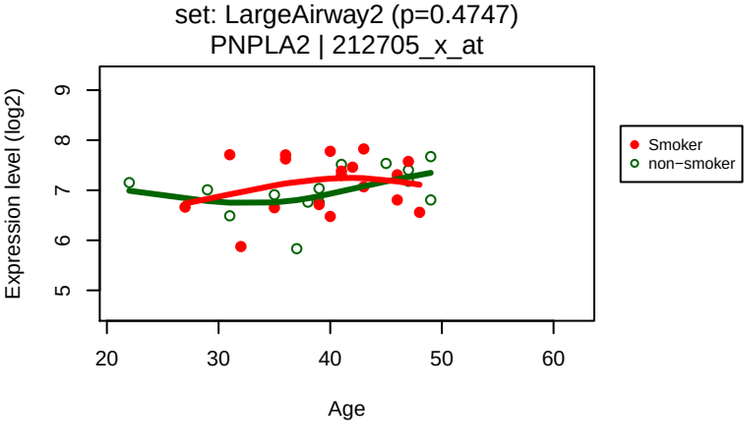

Gene expression ratio (smokers/nonsmokers)
gene: PNPLA2 | 212705_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.144
p=0.4495
p=0.4272
p=0.9787
p=0.4747
p=0.7387
p=0.4583
p=0.2389
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

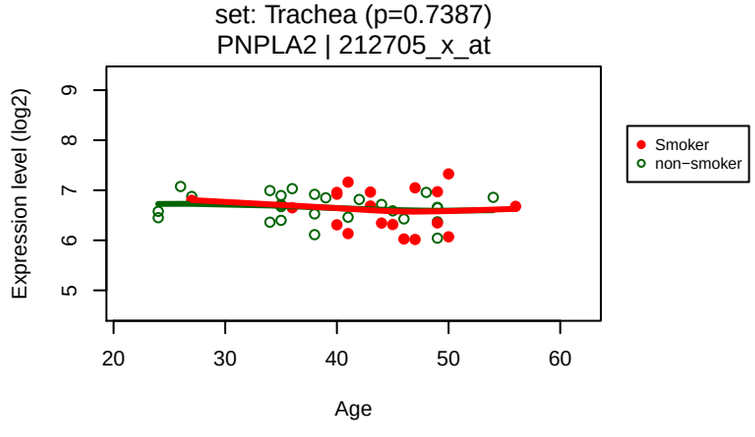

Gene expression ratio in Males
gene: PNPLA2 | 212705_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9532
p=0.0543
p=0.3301
p=0.1767
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PNPLA2 | 212705_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3273
p=0.0854
p=0.0568
p=0.7975
SmallAirway2
LargeAirway2
Trachea
Macrophage2

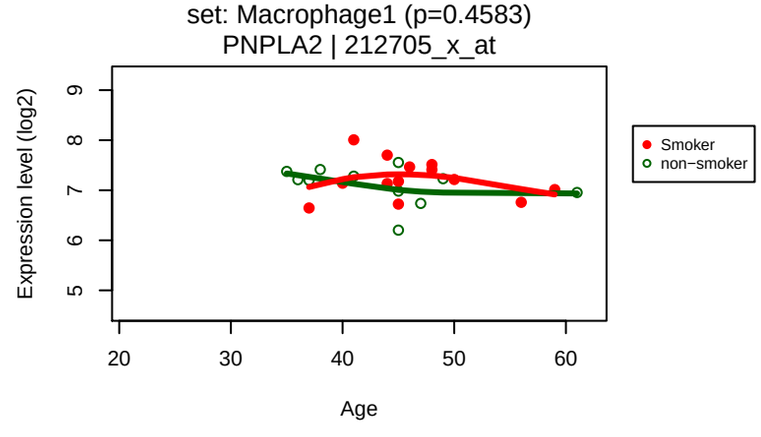

set: SmallAirway1 (p=0.144)
PNPLA2 | 212705_x_at
Expression level (log2)
Age
Smoker
non−smoker

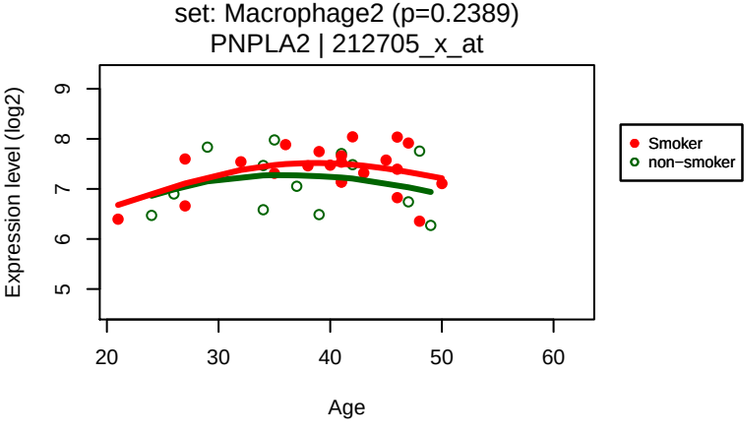

set: SmallAirway2 (p=0.4495)
PNPLA2 | 212705_x_at
Expression level (log2)
Age
Smoker
non−smoker

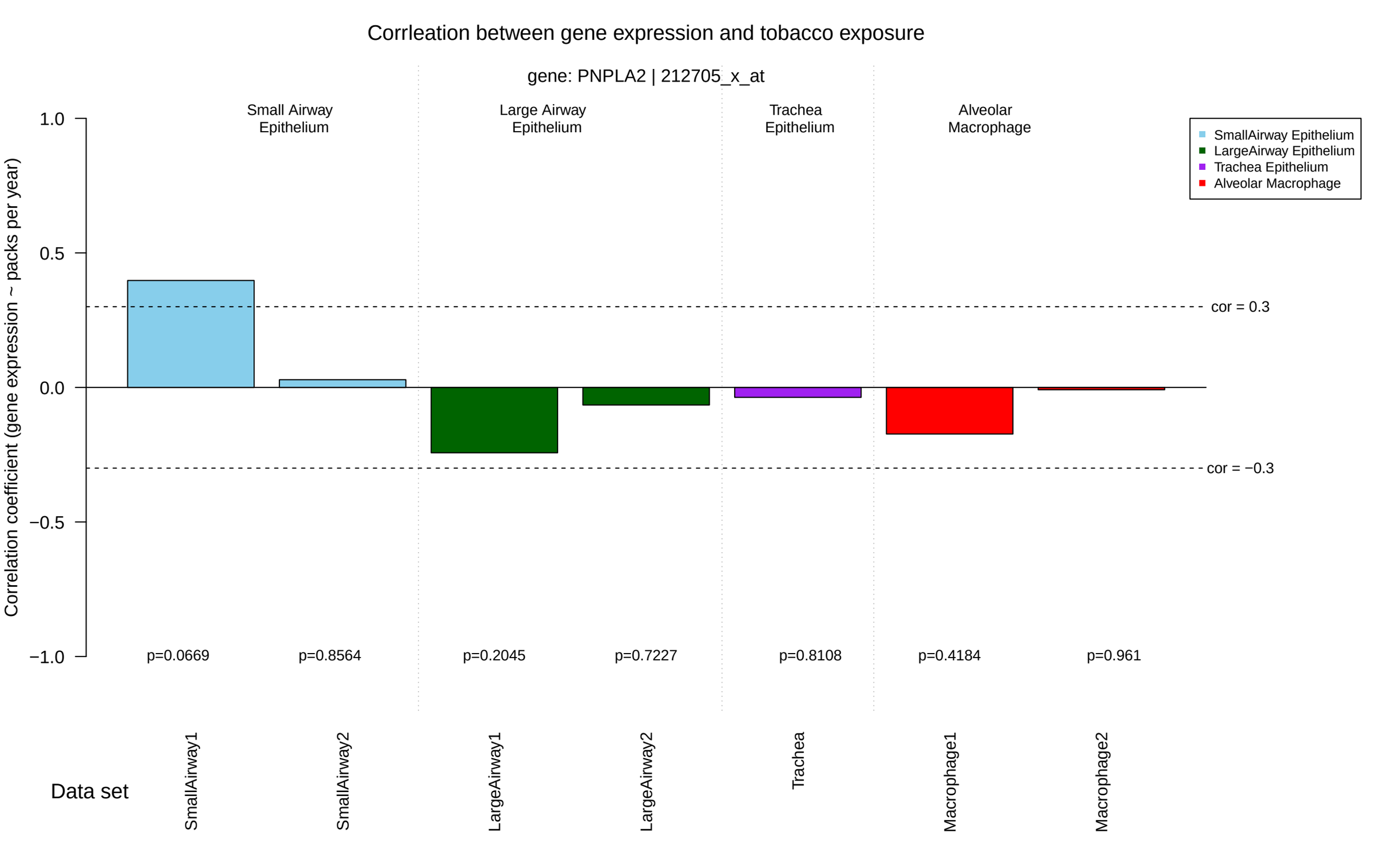

set: LargeAirway1 (p=0.9787)
PNPLA2 | 212705_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4747)
PNPLA2 | 212705_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7387)
PNPLA2 | 212705_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4583)
PNPLA2 | 212705_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2389)
PNPLA2 | 212705_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PNPLA2 | 212705_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0669
p=0.8564
p=0.2045
p=0.7227
p=0.8108
p=0.4184
p=0.961
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2